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Research paper

3D constitutive modeling of the biaxial mechanical response of intact and layer-dissected human carotid arteries

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ABSTRACT

Human arteries with non-atherosclerotic intimal thickening consist of three distinct layers: adventitia, media and intima. From a series of axial extension and inflation tests on intact and layer-dissected human carotid arteries (adventitia and media–intima composite), a 3D structurally-based strain-energy function (SEF) is calibrated, and a set of five material parameters is identified which is not yet available in the literature. The zero-stress states of the artery tubes investigated are considered in the calibration process, and the related kinematics for the finite deformation of the individual continuum are described in detail. The SEF employed is capable of describing the different mechanical properties of the intact and layer-dissected tissue tubes (arterial segments) investigated at different pressure domains and axial stretches. The correlation coefficients and error measures determined indicate good correlation between the model and the experimental data for all tested tubes. Mean values of each individual material parameter provide a kind of ‘master model’ that characterizes the mean response of all mechanical data obtained from the human carotid arteries. The material parameters and the 3D constitutive model serve as a foundation for finite element simulations, and hence the analysis of more complex patient-specific boundary-value problems in the human carotid physiology and pathology.

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1. Introduction

Carotid arteries are of particular clinical and biomedical interest since they are especially prone to atherosclerosis and, therefore, frequently undergo treatments with the aim of reducing the possibility of stroke (Coward et al., 2006). Minimally-invasive techniques, especially carotid angioplasty stenting (CAS) are mechanically based procedures which are becoming increasingly relevant due to the numerous advantages they offer (Yadav et al., 2004; Coward et al.,

2006). Moreover, the mechanical quantities stress and strain are known to trigger the onset of several diseases such as atherosclerosis (Cheng et al., 1993; Richardson, 2002). These facts are some of the reasons why the mechanical properties of arterial wall tissues have received increasing attention in the literature over the past few years (Holzapfel and Ogden, 2010). Generally, for the description of the highly nonlinear and anisotropic mechanics of human arterial tissues under different loading conditions, rationally-based constitutive models are essential. Such constitutive models together with

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available computational (finite element) methods can be used in the optimization process and for the prediction of the outcome of CAS (Wu et al., 2007). Moreover, the origin and the progression of cardiovascular diseases like atherosclerosis, as well as the design and materials of artificial vascular grafts can be investigated with such techniques (Younis et al., 2004).

A key factor for obtaining meaningful results from numerical simulations is the quality of the experimentally-determined mechanical properties of the tissues. However, very little *in vitro* data on the (passive) mechanical properties of human carotid arteries are available in the literature. Uniaxial tensile tests on common carotid intima strips have been performed by Stemper et al. (2005). More importantly, multiaxial *in vitro* tests such as extension–inflation tests on human common carotids have been utilized by Gupta and Kasyanov (1997), Kasyanov et al. (2003) and Guinea et al. (2005). Although the common and internal carotid arteries are known to be the main locations for atherosclerotic plaques that cause cerebrovascular disease, there is only one *in vitro* extension–inflation study available that deals with the determination of the mechanical responses of intact walls and specific layers of these arteries (Sommer et al., 2010). Those experiments show that the passive carotid tissues display elastic, highly nonlinear and anisotropic mechanical behavior under finite strains. Furthermore, carotid tissues show pronounced residual-stress releases in the circumferential and axial directions. After layer-dissection of the intact wall tube, the individual layers also show contraction and elongation in the circumferential and axial directions, respectively. It is of crucial importance to identify these residual stresses in order to reliably predict the state of stress in the arterial tissue (Chuong and Fung, 1986; Takamizawa and Hayashi, 1987; Holzapfel et al., 2000).

Constitutive models based on the theory of incompressible nonlinear hyperelasticity, i.e. the constitutive equations are described in terms of a strain-energy function (SEF) per unit reference volume, are commonly used to describe the mechanical properties of soft biological tissues such as arteries (Holzapfel, 2000; Holzapfel and Ogden, 2010). In the past, most SEFs which were able to describe the typical mechanical characteristics of arterial walls (i.e. nonlinearity, anisotropy and finite strains), were phenomenologically based (Demiray, 1972; Fung et al., 1979; Chuong and Fung, 1983; Takamizawa and Hayashi, 1987). Even though these SEFs succeeded in describing the (macroscopic) material response of various soft biological tissues, most of them have significant limitations, for example, they incorporate too many material parameters, no structural parameters or are oversimplified. Ideally, a SEF should consider histological and structural information for a better description of wall deformations under loading. Appropriate arterial SEFs incorporating the histological structure are discussed in, for example, Hayashi (1993), Humphrey (1999, 2003), Holzapfel et al. (2000, 2005), Zulliger et al. (2004), Gasser et al. (2006) and Holzapfel and Ogden (2010). Structural constitutive models that incorporate the collagen fiber orientation and dispersion are proposed by Holzapfel et al. (2005) and Gasser et al. (2006), which are extensions of the material model by Holzapfel et al. (2000).

In the present study, the authors developed a theoretical framework based on the fiber-dispersion model by Holzapfel

et al. (2005), which is not only able to incorporate the mechanical behavior but also to incorporate the kinematics (residual stretches and curvatures), by using the data from Sommer et al. (2010) of aged non-atherosclerotic human carotid arteries. The advantage of the proposed theoretical framework is that, the resulting constitutive equations for the carotid tissues incorporate the important residual deformations. To the authors' knowledge this is the first study where constitutive equations of comprehensive extension–inflation test data of the intact wall and the individual layers (adventitia and media-intima) for both human common and internal carotid arteries are provided.

2. Materials and methods

2.1. Material

Eleven human common carotid arteries and 10 human internal carotid arteries from 11 human subjects (76.8 ± 6.3 yrs, mean $\pm$ SD, ranging from 67 to 86 yrs, 4 female and 7 male) were excised during autopsy within 24 h from death. The arteries were required to have at most a medium atherosclerotic plaque formation (see Anamnesis Table 1 in Sommer et al., 2010). The plaque grade estimations were made under an examination by experienced pathologists. The use of autopsy material from human subjects was approved by the Ethics Committee of the Medical University of Graz in Austria.

2.2. Mechanical testing

Extension–inflation tests, with continuous recording of axial force F_z , outer diameter and gage length of the intact artery walls were performed at transmural internal pressures p_i ranging from 0 to 33 kPa (250 mmHg) at several axial stretches λ_z ranging from 1.0 to 1.3, in increments of 0.05. The specimens were immersed in a 37 °C calcium-free 0.9% NaCl physiological saline solution during testing.

Thereafter, the intact artery wall was dissected into an adventitia and a media-intima tube, which were tested separately. For the adventitia, the same testing protocol as for the intact artery wall was used ($p_i = 0$ –33 kPa), and then the tests were repeated for a higher pressure range ($p_i = 0$ –100 kPa). Only a cyclic pressurization of 20 kPa was applied to the media-intima tube, because of its inherently low rupture pressure identified in preliminary tests. The circumferential and axial residual stretches of the adventitia and the media-intima tubes were identified photogrammetrically, since the adventitia showed an axial contraction after dissection from the media-intima tube. In order to identify the (nearly) stress-free configurations of the intact artery wall, strips in the axial direction and rings from the adventitia and the media-intima tubes were prepared. Geometrical data such as curvatures, lengths and thicknesses were determined for the intact artery wall, the adventitia and the media-intima. Finally, histological investigations were performed to confirm correct and possibly 'atraumatic' separation of the adventitia from the media-intima composite.

Summary of the testing results. Strong nonlinearity with small hysteresis are common mechanical features of all investigated tissues. The so-called 'inversion'-feature (i.e. a negative slope of the axial stretch–pressure curves at higher

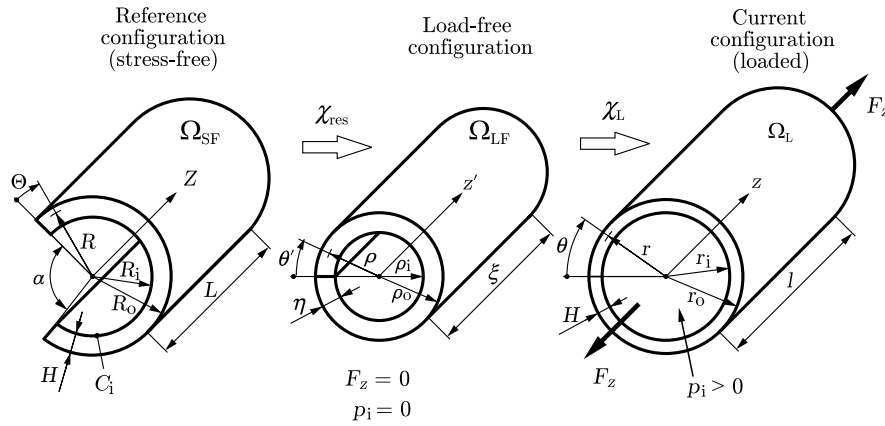


Fig. 1 – Kinematics of the intact wall. An intact arterial ring in the reference (stress-free) configuration Ω_{SF} , with released circumferential residual stresses, the load-free configuration Ω_{LF} with inherent residual stresses but without external loadings, and the current (loaded) configuration Ω_L after the application of internal pressure p_i and axial force F_z .

axial stretches) was observed only for the intact walls. Adventitias were very compliant at low pressures but carried significant loads under physiological conditions. The adventitias became very stiff tubes at pressures exceeding the physiological range. Stress analyses indicated highly anisotropic material properties of all investigated tissues. In addition, high circumferential and axial stresses occurred in the media-intima tube at physiological conditions. For a more detailed description of the testing method employed and the testing results, the reader is referred to Sommer et al. (2010).

2.3. Theoretical framework

2.3.1. Kinematics of an intact arterial wall tube: combined bending, extension and inflation

The intact artery is considered as an incompressible thick-walled cylindrical tube subjected to extension and inflation and we assume that there is no torsion applied on the tube. It is known that the load-free configuration, denoted as Ω_{LF} , in which the artery is excised from the body and not subjected to any loads is not a stress free (or strain free) reference configuration, denoted as Ω_{SF} (Vaishnav and Vossoughi, 1983). Thus, the arterial ring springs open when radially cut and an axially oriented strip bends away from the vessel axis (Holzapfel et al., 2007; Sommer et al., 2010). We assume that the open sector is the stress-free (and fixed) reference configuration Ω_{SF} , as depicted in Fig. 1. Thus, in terms of cylindrical coordinates (R, Θ, Z) , the geometrical region of the stress-free configuration Ω_{SF} of the tube is defined as

$$R_i \leq R \leq R_o, \quad 0 \leq \Theta \leq (2\pi - \alpha), \quad 0 \leq Z \leq L, \quad (1)$$

where R_i , R_o , α and L denote the inner and outer radii, the opening angle and the length of the ‘stress free’ split tube, respectively. Note that the opening angle α identified in Fig. 1 differs from the definition normally used. For example, in Fung and Liu (1989), the opening angle is defined as an angle between two drawn radii from the midpoint of the arc of the inner vessel wall to the outer tips of the open sector. Furthermore, the wall thickness in that configuration Ω_{SF} is denoted as H .

The deformation χ takes Ω_{SF} into the current (loaded) configuration Ω_L , where χ is the composition of the deformations χ_L and χ_{res} , as indicated in Fig. 1. The deformation χ_{res} generates the load-free configuration Ω_{LF} by bending, which is associated with residual stresses, while χ_L is associated with axial elongation and inflation, and leads to the final (current) configuration Ω_L . Particularly, the deformation χ_{res} maps the stress-free configuration Ω_{SF} into the load-free configuration Ω_{LF} . In terms of the cylindrical coordinates (ρ, θ', z') , the region of Ω_{LF} is

$$\rho_i \leq \rho \leq \rho_o, \quad 0 \leq \theta' \leq 2\pi, \quad 0 \leq z' \leq \xi, \quad (2)$$

where ρ_i , ρ_o and ξ denote the inner and outer radii, and the length of the unloaded configuration Ω_{LF} , respectively.

Due to the incompressibility condition, the inner radius of the load-free tube ρ_i is given by

$$\rho_i = \sqrt{\rho_o^2 - \frac{R_o^2 - R_i^2}{k\lambda_{z,res}}}, \quad \theta' = k\Theta, \quad z' = \lambda_{z,res}Z, \quad (3)$$

with

$$k = \frac{2\pi R_i}{C_i} = \frac{2\pi}{\kappa_{R_i} C_i}, \quad (4)$$

where the parameter k , defined by $k = 2\pi/(2\pi - \alpha)$, is a convenient measure of the opening angle α in the stress-free configuration (Holzapfel et al., 2000). The parameter k can be calculated from the inner curvature $\kappa_{R_i} = 1/R_i$ and the inner arc length (circumference) $C_i = R_i(2\pi - \alpha)$, which are measured from photographs (see Holzapfel et al., 2007 and Sommer et al., 2010). The axial stretch $\lambda_{z,res}$ is given by the ratio ξ_i/L_i , where ξ_i and L_i are the lengths of the unloaded inner tube and the inner portion of the excised stress-free axial strip, respectively. Both lengths ξ_i and L_i are identified from axial residual stress (deformation) measurements (see Holzapfel et al., 2007), and are summarized in Table 3 in Sommer et al., 2010. The wall thickness in configuration Ω_{LF} is denoted as η .

Fig. 2 shows the kinematics of the reference configuration Ω_{SF} of an axial strip, which is mapped into the load-free

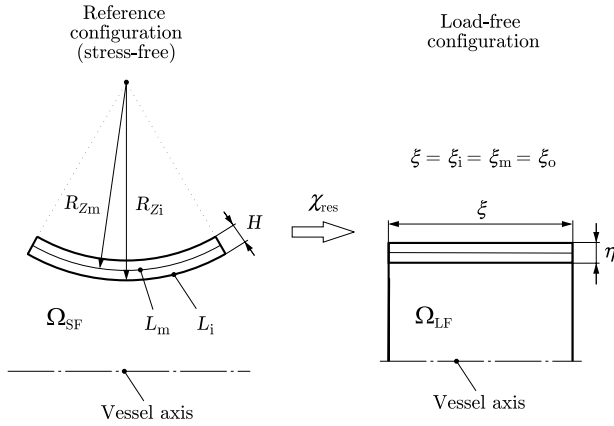


Fig. 2 – Kinematics of the reference configuration Ω_{SF} of an axial strip, assumed to be stress free after residual stress release, mapped into the load-free configuration Ω_{LF} , where the axial strip is constrained to be straight in the composite arterial (layer) ring, see also Fig. 2 in Holzapfel et al. (2007).

configuration Ω_{LF} . For the ratio of the lengths and radii, we can derive from the left part of Fig. 2 that

$$\frac{L_m}{R_{Zm}} = \frac{L_i}{R_{Zi}}, \quad (5)$$

where L_m and L_i are the arc lengths of the middle and inner portions of the stress-free axial strip, respectively, and R_{Zm} and R_{Zi} are the corresponding middle and inner arc radii of the stress-free axial strip, respectively. In addition, with $R_{Zm} = R_{Zi} - H/2$ and $R_{Zi} = 1/\kappa_{Zi}$, from (5), it follows that

$$L_i = \frac{L_m}{1 - \kappa_{Zi} \frac{H}{2}}, \quad (6)$$

where κ_{Zi} is the inner curvature of the stress-free axial strip, prepared for the axial residual stress measurements (Holzapfel et al., 2007). Given the assumption that $L_m = \xi_m$, where ξ_m is the length of the unloaded middle tube of the excised stress-free axial strip, and $\xi = \xi_i = \xi_m = \xi_o$ (the subscript o stands again for outer), the axial residual stretch $\lambda_{z,res} = \xi_i/L_i$ is given by

$$\lambda_{z,res} = 1 - \kappa_{Zi} \frac{H}{2}. \quad (7)$$

The deformation χ_L takes the load-free configuration Ω_{LF} into the current (loaded) configuration Ω_L , see Fig. 1. In terms of cylindrical coordinates (r, θ, z) , the region of the current configuration is

$$r_i \leq r \leq r_o, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq l, \quad (8)$$

where r_i , r_o and l denote the inner and outer radii and the length of the loaded tube, respectively. The wall thickness in that configuration Ω_L is denoted as h . During the deformation χ_L , the inner radius r_i of the loaded tube follows (again using the incompressibility condition) as

$$r_i = \sqrt{r_o^2 - \frac{\rho_o^2 - \rho_i^2}{\lambda_z}}, \quad (9)$$

where $\lambda_z = l/\xi$ is the (measured) axial stretch of the loaded tube with respect to the configuration Ω_{LF} . Furthermore, for

$\chi = \chi_L \circ \chi_{res}$, it follows with (3) that $r_i^2 = r_o^2 - (R_o^2 - R_i^2)/(\kappa_{z,res}\lambda_z)$. Note that the outer radius of the loaded tube may be expressed as

$$r_o = \rho_o \lambda_\theta, \quad (10)$$

where λ_θ is the (measured) circumferential stretch of the loaded tube with respect to the configuration Ω_{LF} . Finally, r_i can be expressed solely as a function of the (indirectly) measured circumferential stretch λ_θ and the axial stretch λ_z of the tube according to

$$r_i = \sqrt{(\rho_o \lambda_\theta)^2 - \frac{R_o^2 - R_i^2}{\kappa_{z,res}\lambda_z}}, \quad (11)$$

where ρ_o , R_o , R_i , κ and $\lambda_{z,res}$ are measured values.

2.3.2. Kinematics of the adventitia and the media-intima tube

The same procedure used for the intact tube can also be applied for the kinematics of the adventitia and the media-intima tubes. However, when the adventitia is removed from the media-intima tube, residual stretches between these tubes are released, and therefore, have to be taken into account, see Fig. 3. Average circumferential and axial residual stretches of the adventitia and the media-intima tubes are summarized in Table 2 in Sommer et al. (2010). Hence, the map between Ω_{SF} and Ω_{LF} also requires the consideration of the circumferential and the axial residual stretches.

The outer radius of the adventitia and the media-intima tube may be expressed as

$$r_o = \rho_o \lambda_{\theta,rs} \lambda_\theta, \quad (12)$$

where $\lambda_{\theta,rs}$ denotes the circumferential residual stretch computed as ρ_{oA}/ρ_o for the adventitia tube, and as ρ_{oMI}/ρ_o for the media-intima tube. For r_i of the adventitia and the media-intima tube, it follows that

$$r_i = \sqrt{(\rho_o \lambda_{\theta,rs} \lambda_\theta)^2 - \frac{R_o^2 - R_i^2}{\kappa_{z,res}\lambda_{z,rs}\lambda_z}}, \quad (13)$$

where $\lambda_{z,rs}$ denotes the axial residual stretch, which is ξ_A/ξ for the adventitia and ξ_{MI}/ξ for the media-intima.

2.3.3. Equilibrium equation

In the absence of body forces, the equilibrium equation is

$$\text{div } \sigma = 0, \quad (14)$$

where $\text{div}(\bullet)$ denotes the spatial divergence of the spatial field $(\bullet)$. Note that in cylindrical coordinates (r, θ, z) , due to geometrical and constitutive symmetry, the only non-trivial component of (14) is

$$\frac{d\sigma_{rr}}{dr} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0. \quad (15)$$

From this equation and the boundary condition $\sigma_{rr}|_{r=r_o} = 0$ on the outer surface of the tube, the radial Cauchy stress σ_{rr} may be calculated as:

$$\sigma_{rr}(\xi) = \int_{\xi}^{r_o} (\sigma_{rr} - \sigma_{\theta\theta}) \frac{dr}{r}, \quad r_i \leq \xi \leq r_o. \quad (16)$$

The internal pressure $p_i = -\sigma_{rr}|_{r=r_i}$ can be obtained by

$$p_i = \int_{r_i}^{r_o} (\sigma_{\theta\theta} - \sigma_{rr}) \frac{dr}{r}. \quad (17)$$

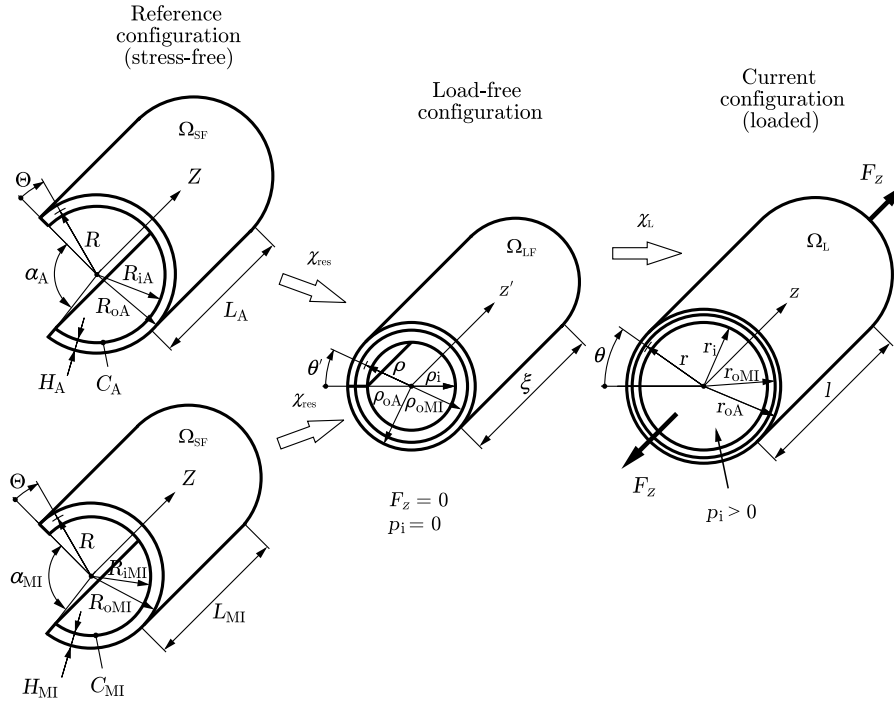


Fig. 3 – Kinematics of the adventitia and the media-intima tubes. Similar to the intact wall, where the two deformations χ_{res} and χ_L are applied to map the (stress-free) reference configuration Ω_{SF} to the load-free configuration Ω_{LF} and to the (loaded) current configuration Ω_L .

In view of the additive split of the Cauchy stress tensor σ into volumetric and isochoric parts, by using the decomposition $\sigma_{\theta\theta} = p + \bar{\sigma}_{\theta\theta}$ and $\sigma_{rr} = p + \bar{\sigma}_{rr}$, where p is the hydrostatic pressure, it follows from (17) that

$$p_i = \int_{r_i}^{r_0} (\bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) \frac{dr}{r}, \quad (18)$$

where $\bar{\sigma}_{\theta\theta}$ and $\bar{\sigma}_{rr}$ denote the isochoric parts of the normal components of the Cauchy stress in the circumferential and radial directions, respectively.

For static equilibrium, the sum of the (measured) axial force F_z and the pressure force $p_i \pi r_i^2$ equals the integral of the axial stress over the vessel wall cross section, i.e.

$$F_z + p_i \pi r_i^2 = 2\pi \int_{r_i}^{r_0} \sigma_{zz} r dr, \quad (19)$$

where σ_{zz} denotes the principal stress in the axial direction. By using the decompositions $\sigma_{zz} = p + \bar{\sigma}_{zz}$, $\sigma_{\theta\theta} = p + \bar{\sigma}_{\theta\theta}$, $\sigma_{rr} = p + \bar{\sigma}_{rr}$, and the expression (17), it follows for the measured force that

$$F_z = \pi \int_{r_i}^{r_0} (2\bar{\sigma}_{zz} - \bar{\sigma}_{\theta\theta} - \bar{\sigma}_{rr}) r dr, \quad (20)$$

where $\bar{\sigma}_{zz}$ denotes the isochoric part of the normal component of the Cauchy stress in the axial direction.

2.3.4. Strain-energy function

In order to model the general mechanical characteristics of the artery tissues, we use the strain-energy function Ψ (per unit volume) as proposed by Holzapfel et al. (2005), i.e.

$$\Psi = \frac{\mu}{2} (I_1 - 3) + \frac{k_1}{k_2} (\exp Q - 1), \quad (21)$$

$$Q = k_2 [(1 - \rho)(I_1 - 3)^2 + \rho(I_4 - 1)^2],$$

with

$$I_1 = \lambda_r^2 + \lambda_\theta^2 + \lambda_z^2, \quad I_4 = \lambda_\theta^2 \cos^2 \varphi + \lambda_z^2 \sin^2 \varphi, \quad (22)$$

where $\mu > 0$ and $k_1 > 0$ are stress-like parameters, $k_2 > 0$ and $\rho \in [0, 1]$ are dimensionless parameters, and I_1 and $I_4 > 1$ are invariants (Holzapfel, 2000). Note that in (21), we actually mean the isochoric (or distortional) elastic response of the material, which, for convenience, we do not particularly mark with bars, a notation introduced above. The fiber families are assumed to be located in the tangential plane of the tube. Consequently, the parameter φ in (22)₂ denotes the angle between the (mean) fiber direction and the circumferential direction in the individual layers, and, therefore, acts as a *geometrical parameter*. Since the structural orientation of the individual layers was not investigated, the parameter φ is used as a phenomenological variable. The dimensionless parameter ρ ranges between zero and one, and it is a measure of the ‘degree of anisotropy’. If $\rho = 0$, the second term of Eq. (21)₁ reduces to an isotropic (rubber-like) model, similar to that proposed by Demiray (1972). If $\rho = 1$, the second term in Eq. (21)₁ reduces to an anisotropic model, as proposed by Holzapfel et al. (2000). Furthermore, ρ can be interpreted as a dispersion (structure) parameter, which represents the three-dimensional fiber distribution around the preferred fiber-direction.

For incompressible hyperelastic materials, the three (isochoric) principal Cauchy stresses $\bar{\sigma}_{rr}$, $\bar{\sigma}_{\theta\theta}$ and $\bar{\sigma}_{zz}$ may be expressed as (Holzapfel, 2000)

$$\bar{\sigma}_{aa} = \lambda_a \frac{\partial \Psi}{\partial \lambda_a}, \quad a = r, \theta, z. \quad (23)$$

Hence, from Eqs. (18) and (23), the strain-energy function (21) and the properties (22) for the invariants I_1 and I_4 , we obtain an explicit expression for the internal pressure p_i , i.e.

$$p_i = 4 \int_{r_i}^{r_o} \left\{ (\lambda_\theta^2 - \lambda_r^2) \left[\frac{\mu}{4} + k_1(1 - \rho)(I_1 - 3) \exp Q \right] + k_1 \rho \lambda_\theta^2 \cos^2 \varphi (I_4 - 1) \exp Q \right\} \frac{1}{r} dr. \quad (24)$$

In an analogous manner, the axial force F_z from (20) can be derived by using Eqs. (21)–(23), thus,

$$F_z = 4\pi \int_{r_i}^{r_o} \left\{ (2\lambda_z^2 - \lambda_\theta^2 - \lambda_r^2) \left[\frac{\mu}{4} + k_1(1 - \rho)(I_1 - 3) \exp Q \right] + k_1 \rho (2\lambda_z^2 \sin^2 \varphi - \lambda_\theta^2 \cos^2 \varphi) (I_4 - 1) \exp Q \right\} r dr. \quad (25)$$

For the intact wall, r_o and r_i are given in (10) and (11), respectively, while for the adventitia and the media-intima tubes, r_o and r_i are given in (12) and (13), respectively. In the absence of shear deformation, the principal stretches of the deformation associated with the radial, circumferential, and axial directions are denoted as λ_r , λ_θ and λ_z , respectively. Each artery tissue sample is regarded as an incompressible material which requires that $\lambda_r \lambda_\theta \lambda_z = 1$, and, therefore, λ_r can be expressed as

$$\lambda_r = (\lambda_\theta \lambda_z)^{-1}. \quad (26)$$

Now with (26), the internal pressure p_i , as provided in (24), and the (measured) axial force F_z according to (25), can be expressed as functions of the (indirectly measured) quantities $\lambda_\theta = r_o/\rho_o$ and $\lambda_z = l/\xi$, and the set $(\mu, k_1, k_2, \varphi, \rho)$ of material parameters, i.e.

$$p_i = p_i(\lambda_\theta, \lambda_z, \mu, k_1, k_2, \varphi, \rho), \quad (27)$$

$$F_z = F_z(\lambda_\theta, \lambda_z, \mu, k_1, k_2, \varphi, \rho).$$

For the intact artery wall, the two functions p_i and F_z contain six constants, which must be identified experimentally, i.e. the radial and axial curvatures κ_{R_i} and κ_{Z_i} , respectively; the inner arc length (circumference) C_i and the wall thickness H in the stress-free configuration Ω_{SF} ; and the outer radius ρ_o and the length ξ in the load-free configuration Ω_{LF} . From these six constants, all other required values can be computed, i.e. $R_i = 1/\kappa_{R_i}$, $k = 2\pi/\kappa_{R_i} C_i$, $R_o = R_i + H$, $\lambda_{z, res} = 1 - H\kappa_{Z_i}/2$, $\lambda_\theta = r_o/\rho_o$ and $\lambda_z = l/\xi$.

Finally, for the adventitia and the media-intima tubes, five additional constants must be identified experimentally: the outer radius R_{oA} and the length Z_A of the adventitia in the stress-free configuration Ω_{SF} , and the corresponding values for the media-intima tube, denoted as R_{oMI} and Z_{MI} , respectively; and finally the outer radius ρ_{oMI} of the unloaded media-intima tube.

2.3.5. Determination of the material parameters

The five material parameters $(\mu, k_1, k_2, \varphi, \rho)$ in (21) and (22) are obtained by means of a nonlinear least-square regression (Levenberg–Marquardt algorithm; Marquardt, 1963). We minimize the objective function

$$E^2 = \sum_{j=1}^n \left\{ \left[w_p (p_{ij}^{\text{mod}} - p_{ij}^{\text{exp}}) \right]^2 + \left[w_F (F_{zj}^{\text{mod}} - F_{zj}^{\text{exp}}) \right]^2 \right\}, \quad (28)$$

where n is the number of data points considered, and w_p and w_F are weighting factors for the internal pressure p_i and the axial force F_z , respectively. p_{ij}^{mod} and F_{zj}^{mod} are the internal pressure and the axial force predicted by the function ψ for the j -th data record ('mod' stands for model), and p_{ij}^{exp} and F_{zj}^{exp} are the associated experimentally measured internal pressure and axial force ('exp' stands for experiment), respectively.

The weighting factors are determined to balance the contributions of the internal pressure and the axial force in the objective function E^2 . Therefore, the mean value of the axial forces is used as weighting factor w_p and the mean value of the internal pressures is used as weighting factor w_F . For the calibration process, 18 representative data points were chosen from the final (i.e. preconditioned) loading cycle of each test data set. Only data points on the loading path were considered. This is justified by the (very) small hysteresis between loading and unloading paths observed for all artery tissues investigated (see the Figs. 6–11 in Sommer et al., 2010).

2.3.6. Statistical analysis

As a measure of the correlation between the model and the experimentally given values for the internal pressure and the axial force, the (linear) correlation coefficients $r(p_i)$ and $r(F_z)$ were calculated as

$$r(p_i) = \frac{\sum_{j=1}^n (p_{ij}^{\text{exp}} - \overline{p_i^{\text{exp}}})(p_{ij}^{\text{mod}} - \overline{p_i^{\text{mod}}})}{\sqrt{\sum_{j=1}^n (p_{ij}^{\text{exp}} - \overline{p_i^{\text{exp}}})^2 \sum_{j=1}^n (p_{ij}^{\text{mod}} - \overline{p_i^{\text{mod}}})^2}}, \quad (29)$$

and

$$r(F_z) = \frac{\sum_{j=1}^n (F_{zj}^{\text{exp}} - \overline{F_z^{\text{exp}}})(F_{zj}^{\text{mod}} - \overline{F_z^{\text{mod}}})}{\sqrt{\sum_{j=1}^n (F_{zj}^{\text{exp}} - \overline{F_z^{\text{exp}}})^2 \sum_{j=1}^n (F_{zj}^{\text{mod}} - \overline{F_z^{\text{mod}}})^2}}, \quad (30)$$

where an overbar denotes an average value.

The correlation coefficients measure the degree of linear correlation between measured internal pressures and axial forces, and those predicted from the strain-energy function (model). Note, however, that even a total correlation of $r = 1$ may be afflicted with poor calibrations. For example, there could be a large constant difference between the associated internal pressure or axial force values resulting in a total correlation of $r = 1$. Therefore, a second error measure, which is based on the objective function E^2 , as given in (28), was calculated to evaluate the 'goodness of fit'. These error measures for p_i and F_z are defined as

$$\varepsilon_{p_i} = \frac{100}{p_{i, \text{ref}}} \sqrt{\frac{E_{p_i}^2}{n - q}}, \quad \varepsilon_{F_z} = \frac{100}{F_{z, \text{ref}}} \sqrt{\frac{E_{F_z}^2}{n - q}}, \quad (31)$$

where $E_{p_i}^2$ and $E_{F_z}^2$ are the final objective function values (or weighted mean-squared errors) of the internal pressure and axial force, respectively. The number of data points considered is represented by n , and q is the number of parameters in the strain-energy function (which is five in our case). Hence $n - q$ is the number of degrees of freedom, while $p_{i, \text{ref}}$ and $F_{z, \text{ref}}$ are the mean values of the internal pressure

Table 1 – Material parameters (μ , k_1 , k_2 , φ , ρ) for the strain-energy function (21) obtained by calibrating the experimental data of the intact wall, the adventitia, and the media-intima composite of the common carotid arteries (Sommer et al., 2010). Furthermore, donor designations (I–XI), location of the tested artery in the throat (left or right), related correlation coefficients $r(p_i)$ and $r(F_z)$ and error measures ε_{p_i} and ε_{F_z} are included.

Donor	Throat side ^a	Material parameters					r		ε (%)	
		μ (kPa)	k_1 (kPa)	k_2	φ (°)	ρ	p_i	F_z	p_i	F_z
Intact wall										
I	l	70.2	99.9	26.4	11.5	1.0	0.970	0.970	8.8	12.2
II	l	46.0	32.7	12.1	19.9	1.0	0.933	0.992	13.3	4.2
II	r	38.7	9.7	19.6	18.9	0.5	0.987	0.941	5.7	13.3
IV	l	34.2	15.5	199.2	17.1	0.8	0.964	0.922	10.6	13.3
V	l	2.2	14.2	64.5	30.3	1.0	0.973	0.955	8.1	14.4
V	r	28.2	8.0	15.1	10.5	0.6	0.983	0.965	6.5	12.5
VI	l	82.1	43.6	120.1	24.2	0.5	0.964	0.963	9.3	12.4
VI	r	54.8	2.9	20.3	31.2	1.0	0.951	0.918	10.3	14.2
VII	r	73.1	5.1	116.8	14.0	0.7	0.956	0.945	11.6	11.2
VIII	r	54.8	13.0	91.5	26.1	0.9	0.995	0.909	3.3	14.1
IX	r	4.9	68.2	71.8	20.9	0.6	0.961	0.983	9.9	8.6
Mean		44.5	28.4	68.9	20.4	0.8	0.967	0.951	8.9	11.9
SD		26.2	30.8	59.3	7.0	0.2	0.018	0.027	2.8	3.0
Adventitia (33 kPa domain)										
II	l	0.0	875.2	146.3	41.2	0.8	0.913	0.908	15.4	15.6
II	r	70.5	61.2	32.8	15.9	0.9	0.939	0.925	12.6	13.3
V	l	95.4	8.5	328.3	38.2	1.0	0.920	0.947	13.0	11.2
V	r	60.0	1.6	41.5	23.9	0.9	0.973	0.973	9.8	15.1
VI	l	34.7	52.2	88.9	26.6	0.7	0.980	0.995	7.1	10.5
VI	r	61.6	28.7	85.8	21.8	0.6	0.989	0.973	5.9	6.2
VIII	r	95.1	238.8	45.3	42.9	0.9	0.939	0.914	13.8	26.5
Mean		59.6	180.9	109.8	30.1	0.8	0.950	0.948	11.1	12.6
SD		33.8	316.6	104.0	10.6	0.1	0.030	0.034	3.6	3.6
Adventitia (100 kPa domain)										
II	l	129.5	506.8	76.8	38.3	0.8	0.962	0.935	9.6	12.8
II	r	217.3	40.6	11.9	19.7	0.9	0.925	0.886	15.6	16.0
V	r	0.0	110.5	43.9	24.9	0.9	0.922	0.903	13.6	15.1
VI	l	98.7	20.9	49.7	21.0	0.7	0.991	0.960	4.8	11.1
VI	r	120.6	23.5	40.8	19.5	0.7	0.950	0.924	8.4	13.4
VIII	r	184.2	39.4	39.2	19.7	0.9	0.984	0.984	7.6	11.4
Mean		138.2	178.4	48.4	25.9	0.8	0.961	0.938	11.9	16.1
SD		77.1	226.4	22.8	8.7	0.1	0.030	0.036	6.3	7.7
Media–Intima										
II	l	134.6	10.3	18.8	8.8	0.7	0.983	0.881	6.4	14.1
IV	l	106.8	26.5	23.0	3.5	0.9	0.970	0.850	8.5	16.3
V	r	128.9	25.9	21.5	6.8	0.7	0.985	0.894	5.9	13.7
VI	r	118.9	36.1	2.8	8.5	1.0	0.999	0.879	1.9	18.0
Mean		122.3	24.7	16.5	6.9	0.8	0.984	0.876	5.7	15.5
SD		12.2	10.7	9.3	2.4	0.2	0.012	0.019	2.8	2.0

^al ... left; r ... right.

and axial force values, respectively. Thus, the error measures ε_{p_i} and ε_{F_z} are the percentages of the root-mean-square errors (per statistical degree-of-freedom) between the experimental values for the internal pressures and axial forces and the related model values, normalized with respect to $p_{i,\text{ref}}$ and $F_{z,\text{ref}}$, respectively.

3. Results

Material parameters for the intact arteries and their related adventitia and media-intima composite tubes are

summarized in the [Tables 1 and 2](#), respectively. In addition, donor designations (I–XI), location of the tested artery in the throat (left or right), the correlation coefficients $r(p_i)$ and $r(F_z)$ and the error measures ε_{p_i} and ε_{F_z} are provided. To identify one set of material parameters (μ , k_1 , k_2 , φ , ρ), the experimental data corresponding to axial stretches λ_z ranging between 1.0 and 1.15 (in increments of 0.05) were considered in the calibration process. [Figs. 4–9](#) show representative circumferential stretch–pressure and axial stretch–pressure plots, where experimental data are compared with numerical results obtained from the SEF (21). Overall, the parameter ρ ranges between 0.8 and 0.9, and the phenomenologically

Table 2 – Material parameters (μ , k_1 , k_2 , φ , ρ) for the strain-energy function (21) obtained by calibrating the experimental data of the intact wall, the adventitia, and the media-intima composite of the internal carotid arteries (Sommer et al., 2010). Furthermore, donor designations (III–XI), location of the tested artery in the throat (left or right), related correlation coefficients $r(p_i)$ and $r(F_z)$ and error measures ε_{p_i} and ε_{F_z} are included.

Donor	Throat side ^a	Material parameters					r		ε (%)	
		μ (kPa)	k_1 (kPa)	k_2	φ (°)	ρ	p_i	F_z	p_i	F_z
Intact wall										
III	l	43.8	93.6	79.6	17.2	0.7	0.925	0.941	9.4	8.5
IV	l	23.7	7.5	12.2	11.6	0.8	0.919	0.905	9.9	12.3
V	r	24.6	13.4	29.9	7.3	0.6	0.951	0.952	10.9	8.9
VI	l	22.9	87.8	109.6	19.3	0.7	0.955	0.980	10.9	10.5
VI	r	3.8	1.2	42.9	19.8	0.7	0.964	0.902	9.1	13.2
VII	r	12.4	93.4	96.9	21.8	0.7	0.970	0.973	8.5	8.3
VIII	r	35.7	1.3	63.1	34.6	0.9	0.947	0.950	11.4	15.6
IX	r	31.8	39.8	80.9	21.3	0.9	0.930	0.910	11.6	13.3
X	r	25.6	20.4	68.7	26.8	0.7	0.950	0.940	10.5	11.7
XI	r	7.4	4.5	45.7	22.4	0.8	0.962	0.904	9.5	15.7
Mean		29.7	27.8	64.2	22.0	0.8	0.944	0.929	10.1	12.3
SD		24.1	36.6	32.6	9.1	0.1	0.019	0.026	1.0	3.0
Adventitia (33 kPa domain)										
VI	r	0.0	15.9	75.5	40.7	1.0	0.926	0.897	13.1	17.8
VII	r	0.3	51.5	38.9	20.4	0.7	0.972	0.905	8.2	13.1
VIII	r	11.3	475.3	63.2	41.6	0.9	0.936	0.930	13.8	15.7
IX	l	3.0	43.2	44.9	29.8	0.8	0.950	0.920	13.4	17.6
X	r	5.7	17.0	76.7	41.3	1.0	0.934	0.912	14.6	18.9
XI	r	149.5	69.9	304.6	16.8	1.0	0.956	0.909	12.3	15.7
Mean		28.3	112.1	100.6	31.8	0.9	0.946	0.912	12.6	16.5
SD		59.5	179.1	101.1	11.2	0.1	0.017	0.012	2.3	2.1
Adventitia (100 kPa domain)										
VI	r	0.0	16.4	55.6	22.4	0.9	0.951	0.890	11.0	19.6
VII	r	0.0	48.2	35.5	29.8	0.7	0.990	0.914	4.9	15.0
VIII	r	0.0	526.6	346.5	30.5	0.9	0.913	0.873	14.9	17.8
X	r	0.0	29.2	48.3	45.4	0.9	0.928	0.908	12.3	15.6
XI	r	0.7	659.5	4.0	0.0	0.8	0.926	0.895	10.4	14.8
Mean		0.1	256.0	98.0	25.6	0.8	0.942	0.896	10.7	16.6
SD		0.3	311.5	140.3	16.6	0.1	0.030	0.016	3.7	2.1
Media-Intima										
IX	r	0.7	17.0	7.6	12.4	0.6	0.999	0.893	1.5	14.8
X	r	36.2	14.2	26.6	10.2	0.8	0.950	0.870	6.2	17.9
XI	r	15.8	32.7	17.8	6.7	0.9	0.932	0.895	8.5	13.7
Mean		17.6	21.3	17.3	9.8	0.8	0.960	0.886	5.4	15.5
SD		17.8	10.3	9.5	2.9	0.2	0.035	0.014	3.6	2.2
^a l ... left; r ... right.										

^al ... left; r ... right.

identified geometrical parameter φ (the fiber-angle) is always smaller than 45° , which reveals (strong) anisotropy and a higher stiffness in the circumferential direction of all tested tubes.

To compare the mechanical behaviors of the different tubes, the circumferential $\sigma_{\theta\theta}$ and the axial σ_{zz} Cauchy stresses are plotted in Figs. 10 and 11 for the three different axial stretches ($\lambda_z = 1.05, 1.1$ and 1.15) against the circumferential stretch λ_θ . We assume an axial *in vivo* stretch of approximately 1.1, i.e. 1.05, 1.1 and 1.15, for the common and internal carotid arteries since the *in vivo* stretch was identified to be 1.07 for a 73 year old male internal carotid artery (Stemper et al., 2005). The plots for the internal carotid arteries look very similar to those of the common carotid arteries and are, therefore, not shown. The plots for the common and internal carotid arteries show a significantly increasing stiffness with

increasing axial stretch, and considerable differences in the mechanical responses between the tubes are observed.

4. Discussion

In this study, it is shown that the three-dimensional strain-energy function (21) provides a suitable model for the extension-inflation experiments performed on aged human common and internal carotid arteries and their layers, as documented in Sommer et al. (2010), see Figs. 4–9 therein. The SEF is suitable to describe the experimental data at different pressure domains and axial stretches: 0–20 kPa for the media-intima composite; 0–33.3 kPa for the intact wall and the adventitia; 0–100 kPa for the adventitia; axial stretches range from 1.0 to 1.3. In particular, the material parameters for the

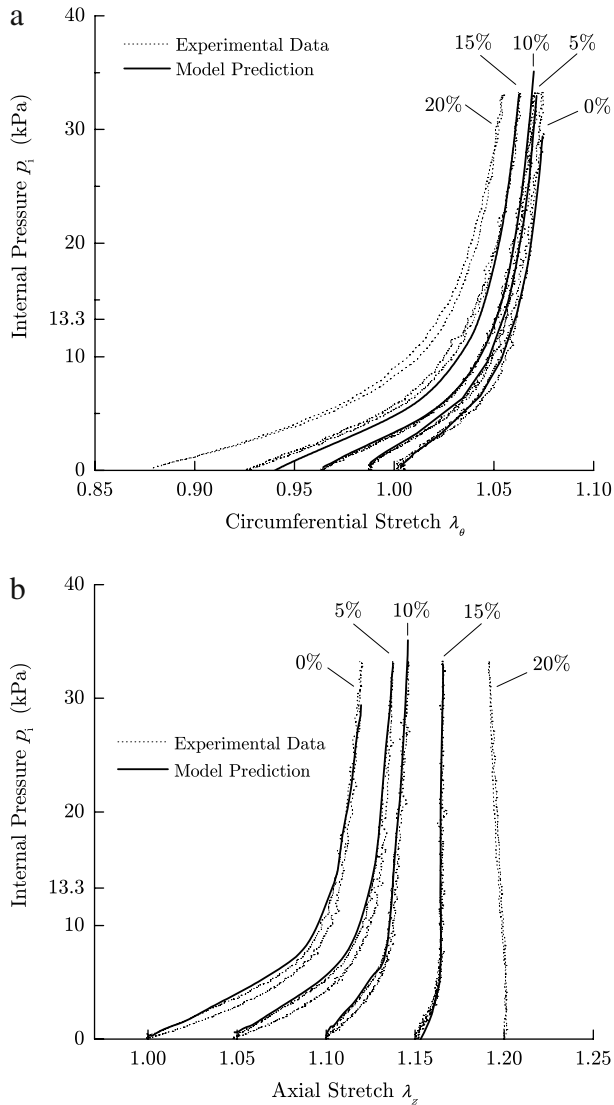


Fig. 4 – Representative plots of pressure–circumferential stretch (a) and pressure–axial stretch (b) for the intact wall of the common carotid artery obtained from donor VIII. Experimental inflation–deflation data (dotted curves) at different axial stretches (ranging from 0 to 20%) are compared with data calculated, predicted by the strain-energy function (21) (solid curves).

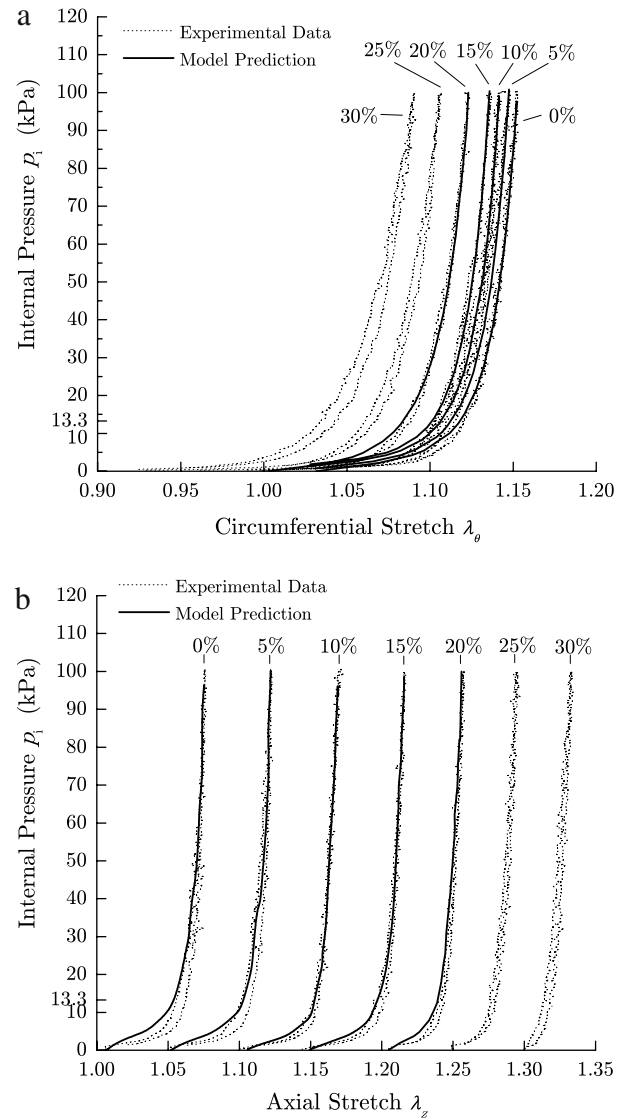


Fig. 5 – Representative plots of pressure–circumferential stretch (a) and pressure–axial stretch (b) for the adventitia of the common carotid artery obtained from donor VIII. Experimental inflation–deflation data (dotted curves) at different axial stretches (ranging from 0 to 30%) are compared with data calculated, predicted by the strain-energy function (21) (solid curves).

3D model of the intact wall, the adventitia and the media-intima composite were identified. For a summary of the parameters, see Tables 1 and 2, and for the representative model calibrations, see Figs. 4–9. As can be seen from these plots, all tubes investigated have strongly nonlinear responses and demonstrate a relatively small hysteresis. At higher axial stretches, only the intact walls show an inversion feature; this means that the tube contracts axially with increasing internal pressure, which is not the case at lower axial stretches. The adventitia is very compliant at lower pressures but change into stiff tubes at higher pressures. The mechanics of the media-intima tubes seem to be almost independent of the applied axial stretches.

Using an appropriate kinematic formulation, it was demonstrated that the material parameters can be identified by considering residual stresses and stretches. Hence, all documented material parameters include the significant (and important) residual stresses and stretches observed in the load-free artery tubes (Holzapfel et al., 2000; Sommer et al., 2010). The 3D model involves only a small number of material parameters so that problems such as non-uniqueness related to the model's sensitivity to small changes in the data (Fung, 1993; Holzapfel et al., 2004) can be avoided within the Levenberg–Marquardt algorithm.

Previous mechanical *in vitro* studies of human carotid arteries, such as the one by Samila and Carter (1981), Gupta

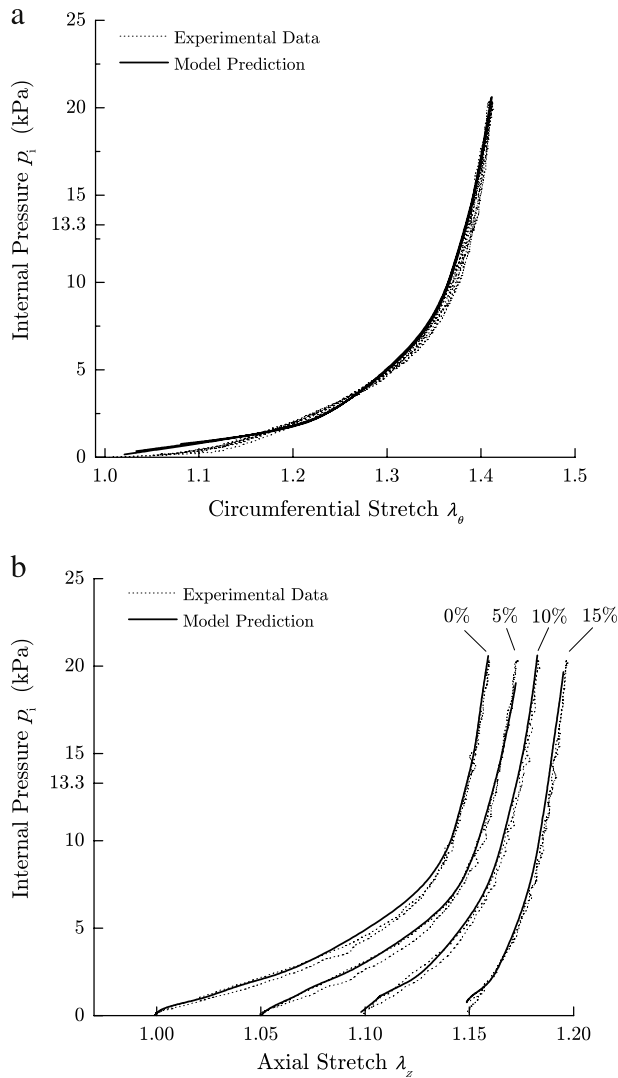


Fig. 6 – Representative plots of pressure–circumferential stretch (a) and pressure–axial stretch (b) for the media-intima of the common carotid artery obtained from donor VI r. Experimental inflation–deflation data (dotted curves) at different axial stretches (ranging from 0 to 15%) are compared with data calculated, predicted by the strain-energy function (21) (solid curves).

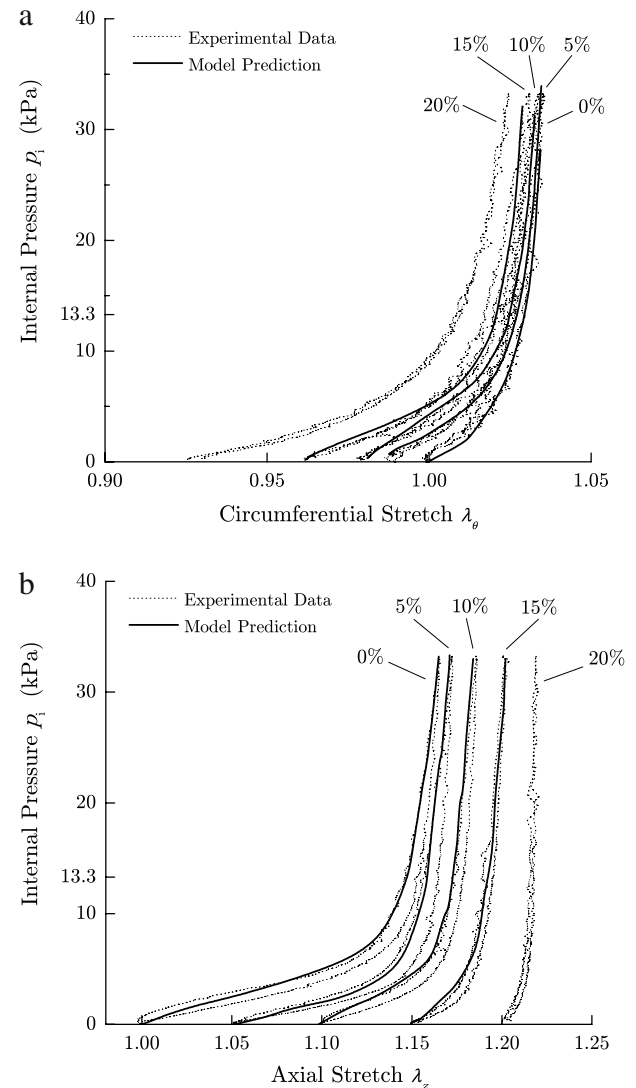


Fig. 7 – Representative plots of pressure–circumferential stretch (a) and pressure–axial stretch (b) for the intact wall of the internal carotid artery obtained from donor VII. Experimental inflation–deflation data (dotted curves) at different axial stretches (ranging from 0 to 15%) are compared with data calculated, predicted by the strain-energy function (21) (solid curves).

and Kasyanov (1997), and Guinea et al. (2005), were not based on experimental data of individual layers. Moreover, there is no constitutive study of human carotid arteries available that focuses on the intact wall and its related layers. The present study seems to be the first which provides material parameters of the intact wall and the individual layers of human common and internal carotid arteries (related data are published by Sommer et al., 2010).

The parameter k_2 is a quantity which may be seen as the average stiffness in the higher loading domain. Interestingly, for every specimen of every layer, the phenomenologically identified fiber angle φ is smaller than 45° , the mean values are even smaller than 32° , which indicates that a higher stiffness is present in the circumferential direction due (most likely) to the collagen fiber orientation. The smallest fiber

angles were observed for the media-intima composite of the common and internal carotid arteries (smaller than 10°), which is consistent with other findings for the medial collagen organization in human arteries (Canham et al., 1991; Finlay et al., 1995). The fiber angle for the adventitia of the common carotid arteries in the 100 kPa domain is always smaller than that of the adventitia in the 33 kPa domain (see Table 1). This phenomenon could be explained by the successive recruitment of collagenous fibers in the outer layers during increased loading, which are initially less organized than those fibers in the inner layers (Schulze-Bauer et al., 2002). This phenomenon was not observed for the internal carotid arteries. For all tested layers, the mean value of the dimensionless (anisotropy) parameter ρ was about 0.8

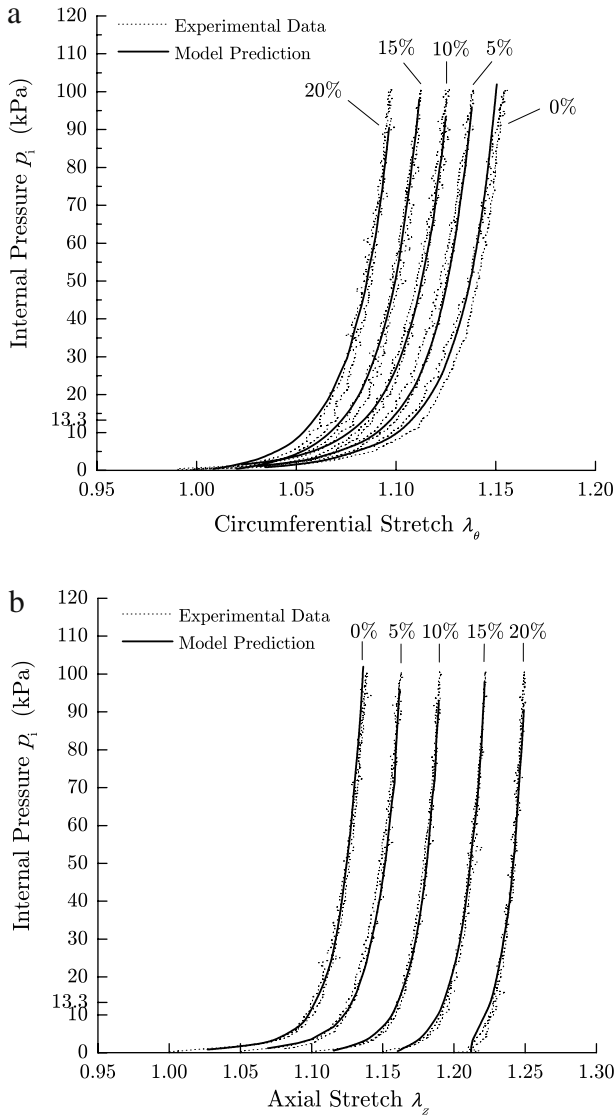


Fig. 8 – Representative plots of pressure–circumferential stretch (a) and pressure–axial stretch (b) for the adventitia of the internal carotid artery obtained from donor VII. Experimental inflation–deflation data (dotted curves) at different axial stretches (ranging from 0 to 20%) are compared with data calculated, predicted by the strain-energy function (21) (solid curves).

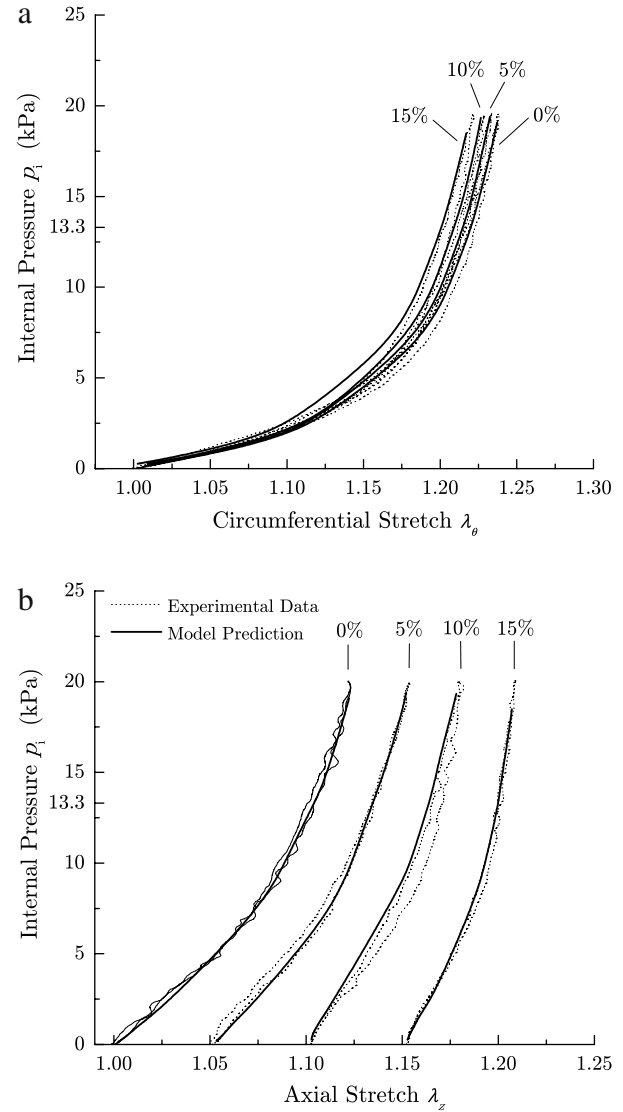


Fig. 9 – Representative plots of pressure–circumferential stretch (a) and pressure–axial stretch (b) for the media-intima of the internal carotid artery obtained from donor IX r. Experimental inflation–deflation data (dotted curves) at different axial stretches (ranging from 0 to 15%) are compared with data calculated, predicted by the strain-energy function (21) (solid curves).

with a small standard deviation of 0.2, which indicates a highly anisotropic material. Both the correlation coefficients (which are close to 1) and the small ‘error measure’ values in Tables 1 and 2, indicate good correlation between the model and the experimental data for all tested tubes, see Figs. 4–9.

By taking the mean value of each individual material parameter, we obtain a kind of ‘master model’, which reflects the mean response of all mechanical data obtained from the carotid arteries. Although the variances among individual specimens are large, the mechanical characteristics become more clear, and this approach allows a comparative study of the mechanics of the different layers investigated. The master model response for the individual layers (illustrated in Figs. 10 and 11) shows that the adventitia stiffens most

rapidly. Whereas, the slope of the stress–stretch relation is larger for the media-intima composite than the intact wall up to a certain stress (stretch) level, and, thereafter, the intact wall becomes stiffer than the media-intima.

The influence of the axial stretch λ_z on the relationship between the Cauchy stress and the circumferential stretch λ_θ for the common carotid artery can also be seen in Figs. 10 and 11. The stress–stretch curves shift significantly to the left, hence becoming stiffer, when the axial stretch increases. This shift was more pronounced in the σ_{zz} – λ_θ plots than in the $\sigma_{\theta\theta}$ – λ_θ plots. A similar behavior was observed for the layers of the internal carotid artery.

The objective function E^2 was minimized to identify the material parameters. In this study, the internal pressure p_i

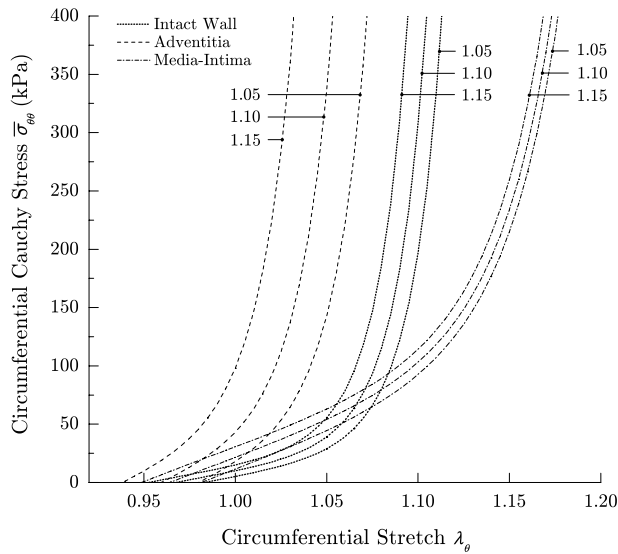


Fig. 10 – Circumferential stress–circumferential stretch model response representing the mean of all mechanical data on the intact wall, the adventitia (33 kPa domain), and the media-intima composite of the common carotid arteries. Mean material parameters are from Table 1. Data correspond to axial stretch ratios $\lambda_z = 1.05, 1.1, 1.15$.

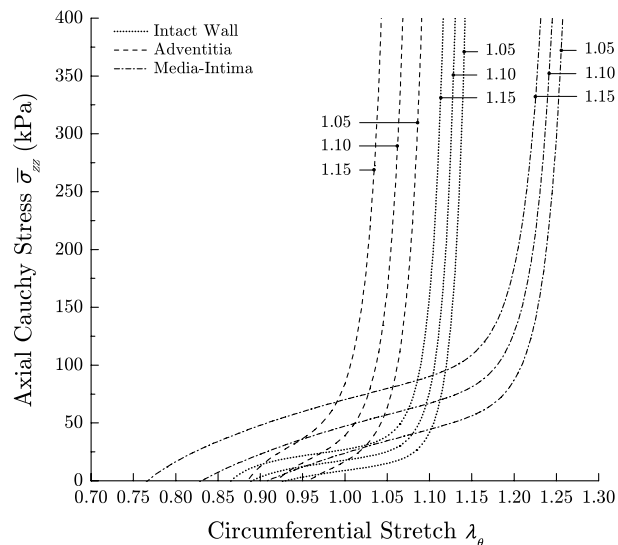


Fig. 11 – Axial stress–circumferential stretch model response representing the mean of all mechanical data on the intact wall, the adventitia (33 kPa domain), and the media-intima composite of the common carotid arteries. Mean material parameters are from Table 1. Data correspond to axial stretch ratios $\lambda_z = 1.05, 1.1, 1.15$.

and the (measured) axial force F_z were used as the governing measures in the objective function. In isometric tests, in which the length of the tube was held constant, we observed a buckling phenomenon of the intact wall and the media-intima tube at higher pressures, and, therefore, distorted axial

stretch measurements. As a consequence, we have let the machine control the external axial force to be approximately constant. Hence, the ‘goodness of fit’ was generally better for the internal pressure p_i than for the axial force F_z (see Tables 1 and 2). The values of the material parameters in (21) and (22) must be such that the strain-energy function Ψ is convex, i.e. the material must be stable under loading (Holzapfel et al., 2000). For validation, we generated contour plots of the strain-energy function with material parameters given in the tables. All contours for all specimens turned out to be convex.

Our approach is based on the assumption that in the ‘stress-free’ configuration, the opened arterial sector in the circumferential and axial directions is stress free. As shown in several studies, every cut leads to a redistribution of residual stresses (see, for example, Volokh, 2006), and an arterial ring cut open is, in general, not stress free. In fact, the body needs to be cut into infinitesimally small parts to become stress free, which will lead to an incompatible reference configuration. The master model parameters stated in this study are only valid for the particular age group from which we harvested the human tissue samples. Patients who undergo medical interventions to prevent stroke, like carotid angioplasty and stenting or carotid endarterectomy, are about this age group. Since different experimental studies have shown that all three layers of an artery become stiffer with age, the model parameters μ , k_1 and k_2 for arterial tissues of younger subjects may be lower than those stated here. The increasing stiffness of aged arteries is assumed to be a consequence of collagen accumulation and elastin degradation. A recent study regarding the collagen fiber orientation and dispersion in the layers of human aortas has not shown any age dependency (Schriebl et al., in press).

The assumption of incompressibility may lead to an underestimation of stresses, because a stress-induced extrusion of the fluid may occur. Moreover, the loose outer adventitial parts detected by the video-extensometer are associated with smaller cross sections and thus higher stresses may occur therein. However, if there is a fluid exchange during cyclic loading, it must occur very rapidly because the hysteresis observed is small (see Figs. 4–9). Only a relatively small number of samples was investigated so that a (meaningful) correlation between the material parameters and the different layers could not be quantified.

Since the (distributed) collagen fiber orientations in the tissues were not identified experimentally, we handled φ and ρ as calibration parameters, and, therefore, as phenomenologically identified parameters. In future works, the use of experimentally identified collagen fiber orientations and associated distributions will lead to more conclusive and reliable material parameters, and therefore, a better description of artery walls; for a recent example of the determination of distributed collagen fiber orientations in human arteries, see the study by Schriebl et al. (in press).

The material parameters and the strain-energy function used in this study are intended to serve as a basis for finite element simulations of clinical intervention methods and related problems, and thus to help to identify the possible improvements better.

REFERENCES

- Canham, P.B., Talman, E.A., Finlay, H.M., Dixon, J.G., 1991. Medial collagen organization in human arteries of the heart and brain by polarized light microscopy. *Connect. Tissue Res.* 26, 121–134.
- Cheng, G.C., Loree, H.M., Kamm, R.D., Fishbein, M.C., Lee, R.T., 1993. Distribution of circumferential stress in ruptured and stable atherosclerotic lesions: a structural analysis with histopathological correlation. *Circulation* 87, 1179–1187.
- Chuong, C.J., Fung, Y.C., 1986. On residual stress in arteries. *J. Biomech. Eng.* 108, 189–192.
- Chuong, C.J., Fung, Y.C., 1983. Three-dimensional stress distribution in arteries. *J. Biomech. Eng.* 105, 268–274.
- Coward, L.J., Featherstone, R.L., Brown, M.M., 2006. Percutaneous transluminal angioplasty and stenting for carotid artery stenosis (review). *The Cochran Database of Systematic Reviews* 2004 2, 1–18.
- Demiray, H., 1972. A note on the elasticity of soft biological tissues. *J. Biomech.* 5, 309–311.
- Finlay, H.M., McCullough, L., Canham, P.B., 1995. Three-dimensional collagen organization of human brain arteries at different transmural pressures. *J. Vasc. Res.* 32, 301–312.
- Fung, Y.C., 1993. *Biomechanics. Mechanical Properties of Living Tissues*, 2nd ed. Springer-Verlag, New York.
- Fung, Y.C., Fronek, K., Patitucci, P., 1979. Pseudoelasticity of arteries and the choice of its mathematical expression. *Am. J. Physiol.* 237, H620–H631.
- Fung, Y.C., Liu, S.Q., 1989. Change of residual strains in arteries due to hypertrophy caused by aortic constrictions. *Circ. Res.* 65, 1340–1349.
- Gasser, T.C., Ogden, R.W., Holzapfel, G.A., 2006. Hyperelastic modelling of arterial layers with distributed collagen fibre orientations. *J. R. Soc. Interface* 3, 15–35.
- Guinea, G.V., Atienza, J.M., Elices, M., Aragoncillo, P., Hayashi, K., 2005. Thermomechanical behavior of human carotid arteries in the passive state. *Am. J. Physiol. Heart Circ. Physiol.* 288, 2940–2945.
- Gupta, B.S., Kasyanov, V.A., 1997. Biomechanics of human common carotid artery and design of novel hybrid textile compliant vascular grafts. *J. Biomed. Mater. Res. A* 24, 341–349.
- Hayashi, K., 1993. Experimental approaches on measuring the mechanical properties and constitutive laws of arterial walls. *J. Biomech. Eng.* 115, 481–488.
- Holzapfel, G.A., 2000. *Nonlinear Solid Mechanics. A Continuum Approach for Engineering*. John Wiley & Sons, Chichester.
- Holzapfel, G.A., Gasser, T.C., Ogden, R.W., 2000. A new constitutive framework for arterial wall mechanics and a comparative study of material models. *J. Elasticity* 61, 1–48.
- Holzapfel, G.A., Gasser, T.C., Ogden, R.W., 2004. Comparison of a multi-layer structural model for arterial walls with a Fung-type model, and issues of material stability. *J. Biomech. Eng.* 126, 264–275.
- Holzapfel, G.A., Ogden, R.W., 2010. Constitutive modelling of arteries. *Proc. R. Soc. Lond. A* 466, 1551–1597.
- Holzapfel, G.A., Sommer, G., Auer, M., Regitnig, P., Ogden, R.W., 2007. Layer-specific 3D residual deformations of human aortas with non-atherosclerotic intimal thickening. *Ann. Biomed. Eng.* 35, 530–545.
- Holzapfel, G.A., Sommer, G., Gasser, C.T., Regitnig, P., 2005. Determination of the layer-specific mechanical properties of human coronary arteries with non-atherosclerotic intimal thickening, and related constitutive modelling. *Am. J. Physiol. Heart Circ. Physiol.* 289, H2048–H2058.
- Humphrey, J.D., 1999. An evaluation of pseudoelastic descriptors used in arterial mechanics. *J. Biomech. Eng.* 121, 259–262.
- Humphrey, J.D., 2003. Continuum biomechanics of soft biological tissues. *Proc. R. Soc. Lond. A* 459, 1–44.
- Kasyanov, V., Ozolanta, I., Purinya, B., Ozols, A., Kancevich, V., 2003. Compliance of a biocomposite vascular tissue in longitudinal and circumferential directions as a basis for creation of artificial substitutes. *Mech. Comp. Mat.* 39, 347–358.
- Marquardt, D.W., 1963. An algorithm for least-squares estimation of nonlinear parameters. *SIAM J. Appl. Math.* 11, 431–441.
- Richardson, P.D., 2002. Biomechanics of plaque rupture: progress, problems, and new frontiers. *Ann. Biomed. Eng.* 30, 524–536.
- Samila, Z.J., Carter, S.A., 1981. The effect of age on the unfolding of elastin lamellae and collagen fibers with stretch in human carotid arteries. *Can. J. Physiol. Pharmacol.* 59, 1050–1057.
- Schriefl, A.J., Zeindlinger, G., Pierce, D.M., Regitnig, P., Holzapfel, G.A., 2011. Determination of the layer-specific distributed collagen fiber orientations in human thoracic and abdominal aortas and common iliac arteries (in press).
- Schulze-Bauer, C.A.J., Regitnig, P., Holzapfel, G.A., 2002. Mechanics of the human femoral adventitia including high-pressure response. *Am. J. Physiol. Heart Circ. Physiol.* 282, H2427–H2440.
- Sommer, G., Regitnig, P., Kolltringer, L., Holzapfel, G.A., 2010. Biaxial mechanical properties of intact and layer-dissected human carotid arteries at physiological and supra-physiological loadings. *Am. J. Physiol. Heart Circ. Physiol.* 298, H898–H912.
- Stemper, B.D., Yoganandan, N., Pintar, F.A., 2005. Methodology to study intimal failure mechanics in human internal carotid arteries. *J. Biomech.* 38, 2491–2496.
- Takamizawa, K., Hayashi, K., 1987. Strain energy density function and uniform strain hypothesis for arterial mechanics. *J. Biomech.* 20, 7–17.
- Vaishnav, R.N., Vossoughi, J., 1983. Estimation of residual strains in aortic segments. In: Hall, C.W. (Ed.), *Biomedical Engineering II: Recent Developments*. Pergamon Press, New York, pp. 330–333.
- Volokh, K.Y., 2006. Stresses in growing soft tissues. *Acta Biomater.* 2, 493–504.
- Wu, W., Qi, M., Liu, X.P., Yang, D.Z., Wang, W.Q., 2007. Delivery and release of nitinol stent in carotid artery and their interactions: a finite element analysis. *J. Biomech.* 40, 3034–3040.
- Yadav, J.S., Wholey, M.H., Kuntz, R.E., Fayad, P., Katzen, B.T., Mishkel, G.J., Bajwa, T.K., Whitlow, P., Strickman, N.E., Jaff, M.R., Popma, J.J., Snead, D., Cutlip, D.E., Firth, B.G., Ouriel, K., 2004. Protected carotid-artery stenting versus endarterectomy in high-risk patients. *N. Engl. J. Med.* 351, 1493–1501.
- Younis, H.F., Kaazempur-Mofrad, M.R., Chan, C., Isasi, A.G., Hinton, D.P., Chau, A.H., Kim, L.A., Kamm, R.D., 2004. Hemodynamics and wall mechanics in human carotid bifurcation and its consequences for atherogenesis: investigation of inter-individual variation. *Biomech. Model. Mechanobiol.* 3, 17–32.
- Zulliger, M.A., Fridez, P., Hayashi, K., Stergiopoulos, N., 2004. A strain energy function for arteries accounting for wall composition and structure. *J. Biomech.* 37, 989–1000.